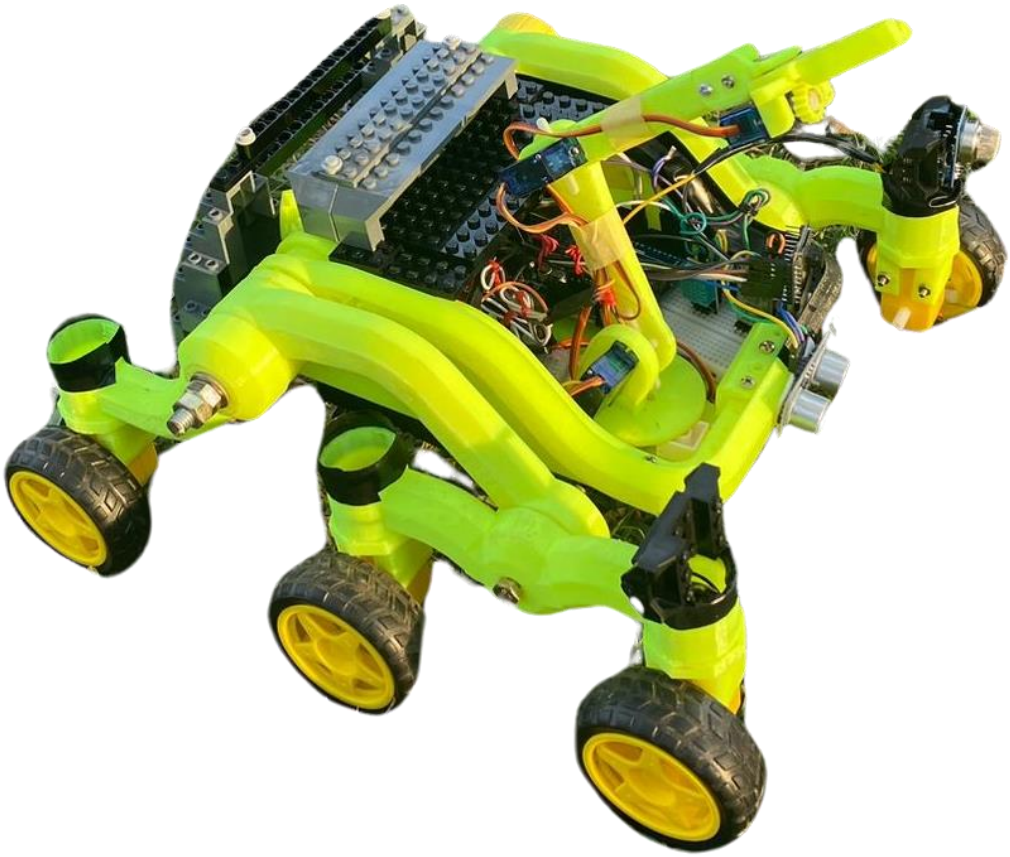
March, 2024



SEEKERBOT

APOLO CONTEST

Project Name:	SEEKERBOT
Topic:	Robótica espacial
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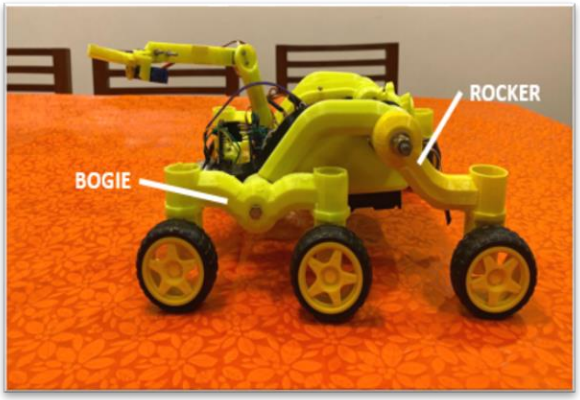
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1. Project description

Our project focuses on the design and construction of a rover capable of navigating across uneven terrain simulating planetary and lunar conditions. The space robot, or rover, *SeekerBot* , will perform functions such as avoiding obstacles, collecting objects from the surface, and gathering environmental information, transmitting it through specialized applications while maintaining wireless control. This proposal integrates knowledge of robotics, Arduino programming, circuit construction, design, and 3D printing.

2. Mechanical Design

2.1. Rover structure



SeekerBot It has a six-wheel design and is equipped with a Rocker-Bogie suspension mechanism . This mechanism is defined as a system composed of two parts: the " rocker" arm " (swing arm) and the " bogie " (balancer). These components are linked together and also have freedom of rotation, allowing the Rover to effectively adapt to uneven terrain, overcoming obstacles and maintaining optimal balance and stability when exploring rocky or sloping areas.

2.2. Robotic arm



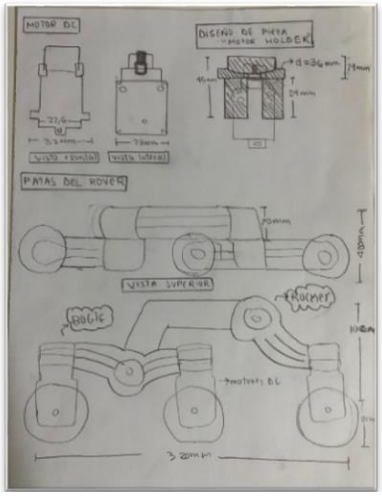
The robotic arm is used to collect surface samples and manipulate objects that interrupt the rover's trajectory. It consists of four main parts, also known as joints: the base, the arm, the forearm, and the claw. The base can rotate the entire structure within a range of 0° to 180°; the arm is responsible for raising and lowering the rover; the forearm controls forward and backward movement; and the

claw is used to release and grasp objects by opening and closing. Each joint is powered by an SG90 servomotor, enabling precise angular control.

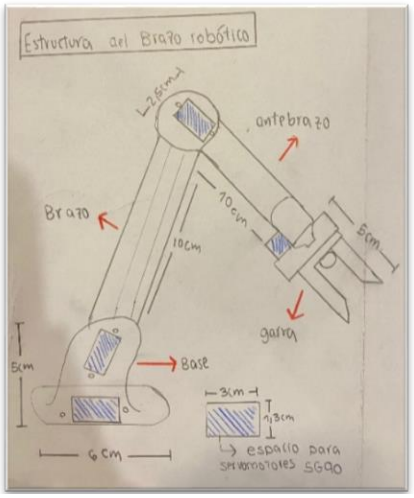
Each piece of the mechanical design was created in software and 3D printed. Similarly, control of both the robotic arm and the rover was made possible through applications created as part of the project. These aspects will be detailed throughout the technical report.

3. Digital sketches and designs

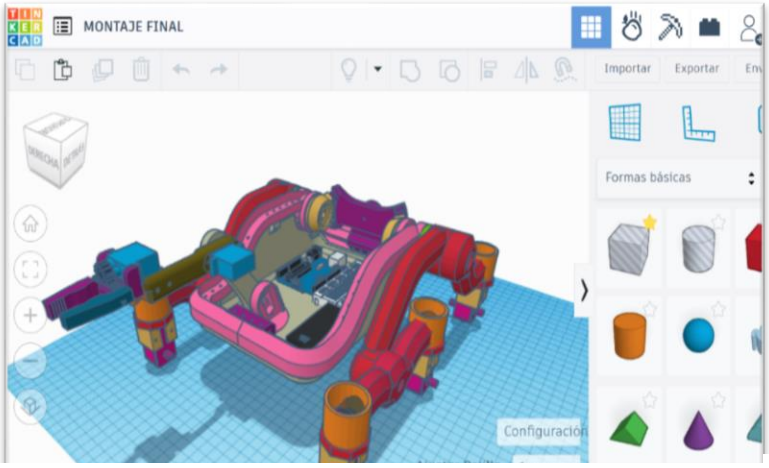
Sketches and paper diagrams were created to visualize the layout and size of each of the SeekerBot 's parts . Each element was then designed using Tinkercad, a free online program created by Autodesk for creating three-dimensional structures. Throughout this process, real-world dimensions were considered, emphasizing the varying thicknesses of the parts and the space required to accommodate the required servomotors, circuit boards, breadboards , and screws.



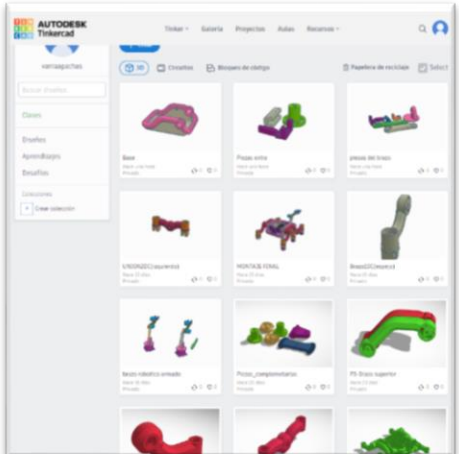
Boceto de extremidades inferiores



Boceto de brazo robótico



Montaje del Rover en Tinkercad

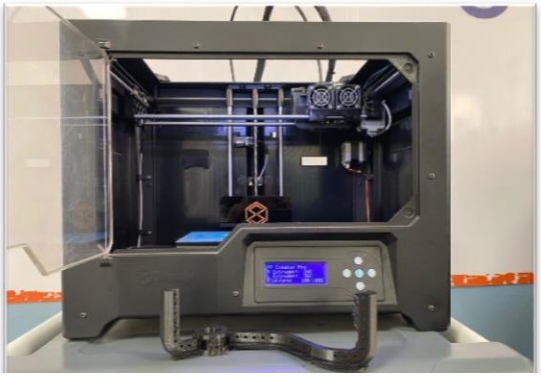


Librería de piezas diseñadas

4. Manufacturing

The Rover's mechanical parts were manufactured using 3D printing, using the Flashforge printer. Creator Pro and PLA (polylactic acid) filaments , selected for their lightness, strength, and excellent finish. The printing process was carried out using designs previously modeled in Tinkercad, which were exported in STL format for the printer to interpret. The estimated total printing time was approximately two weeks, including the time required to reprint parts damaged by filament blockages, adhesion difficulties, and feeding issues.

5. Electronics and programming



Impresora FlashForge Creator pro



Piezas impresas con PLA

5.1. Electronic components

5.1.1. Plates

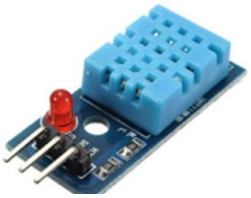
Two types of ARDUINO boards were used in the project: the ARDUINO UNO and the ARDUINO NANO. These boards are based on ATmega328P microcontrollers. The ARDUINO UNO was implemented to control the sensors, while the NANO was implemented to control the robotic arm.

5.1.2. Modules

In this section, the HC-06 and HC-05 Bluetooth modules were incorporated, enabling wireless data transmission and reception. An ESP32-CAM was also integrated for real-time image processing and an L298N motor driver to control the movement of the gear motors. Finally, a NEO-6M GPS module was installed to record longitude and latitude data of the rover's location, which operates via satellite connection.

5.1.3. Sensors

The project incorporated three sensors with different purposes. The DHT-11 sensor was used to obtain temperature and humidity data, the HC-SR04 ultrasonic sensor was used to measure the distance between the rover and an object, and the MQ-135 sensor was used to monitor air quality by detecting gases in the environment. The latter follows the standards established in the Air Quality Index (AQI), which is measured on a scale from 0 to 500.



Sensor DHT11



Sensor HC-SR04



Sensor MQ-135

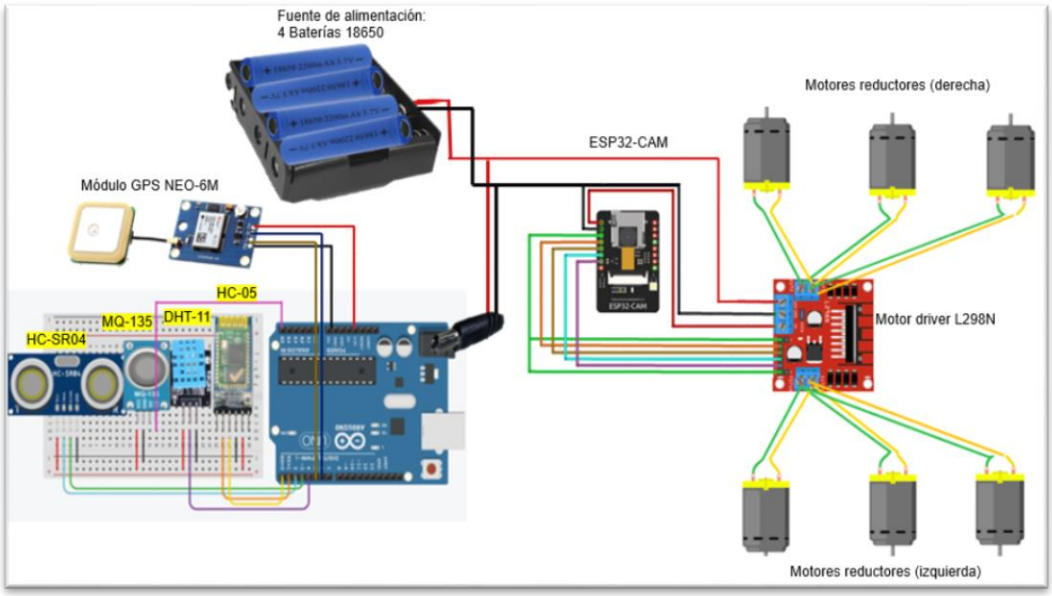
5.2. Circuits

The robot's operation was achieved through two main circuits, each of which accounted for the input and output voltages and power consumption of each module, sensor, and board. Both circuits used 18650 lithium rechargeable batteries, each with a nominal voltage of 3.7V, as a power source.

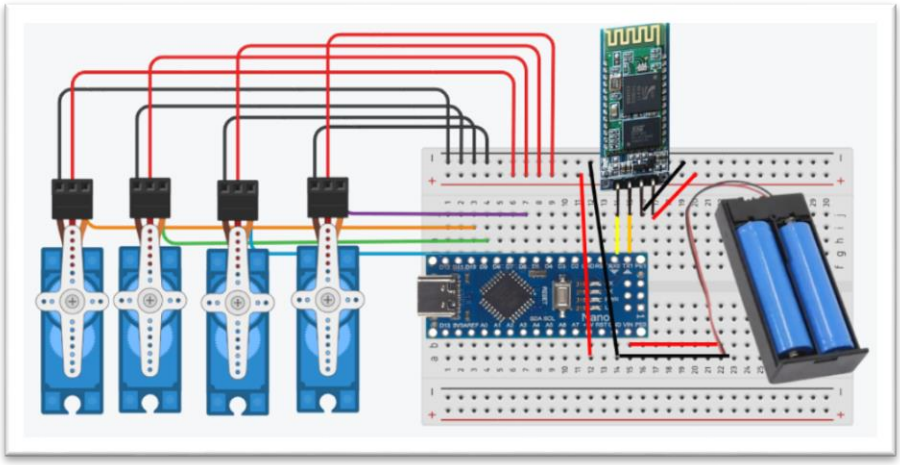
5.2.1. Movement, camera and sensors

This circuit uses four lithium batteries, for a total voltage of 14.8V. These batteries provide 12V to the L298N Motor Driver, which is connected to the six DC motors and the ESP32-CAM. They also power the ARDUINO UNO board, which supplies power to all the sensors and the Bluetooth and GPS modules.

5.2.2. Robotic arm



In this second circuit, the power source consisted of two lithium batteries with a total voltage of 7.4V, connected to the VIN pin of the ARDUINO NANO board. The board, through a breadboard , powered the four SG90 servomotors of the robotic arm and an HC-06 Bluetooth module.

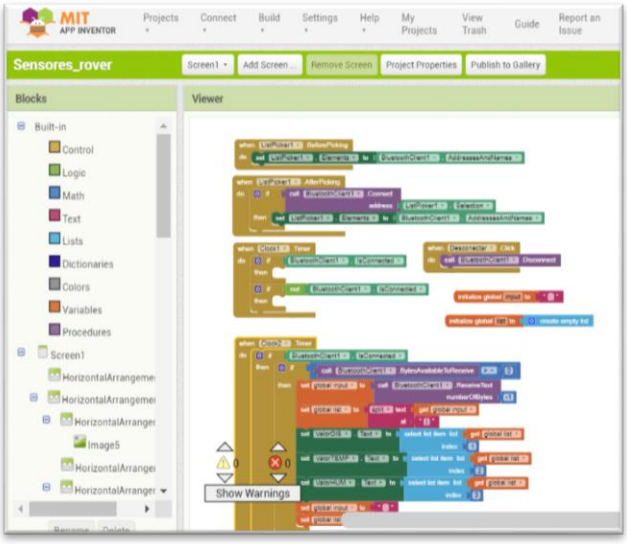


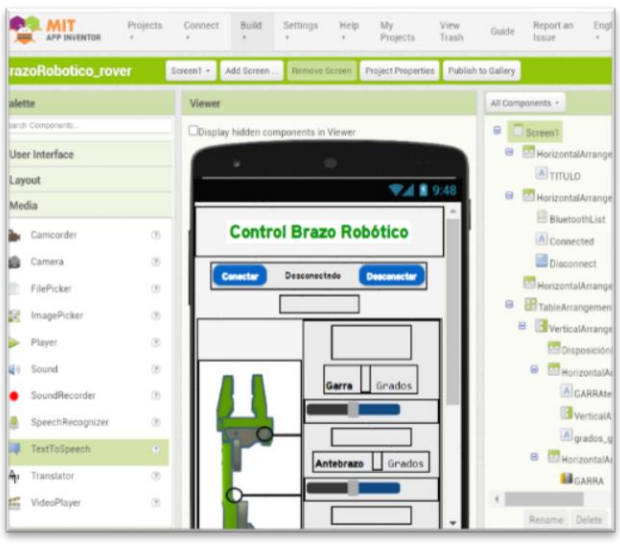
5.3. Programming

SeekerBot is programmed using the Arduino IDE software. This platform allows us to perform various tasks, such as controlling the movements of the DC motors (forward, backward, right, left), creating logic and adjusting the angles of the servo motors for the robotic arm's movement, and printing the data collected by the sensors in sequence.

6. Control and Communication

During the project, we created and designed two applications using the MIT App Inventor software development environment. The first application allowed us to visualize sensor information in real time, such as the distance to an object, temperature, humidity, air quality,

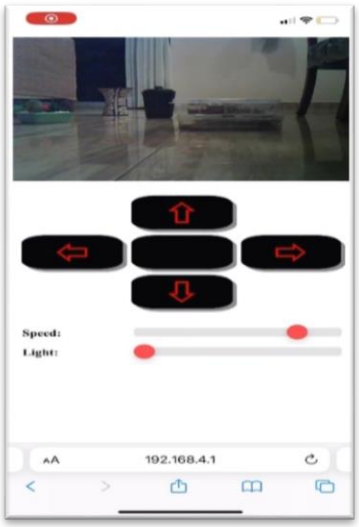




and latitude and longitude coordinates. This application features a simple interface and updates values immediately.

The second application is dedicated to controlling the robotic arm. We use sliders as interface elements to facilitate adjusting the degrees of each servomotor (base, arm, forearm, and gripper). Both can be accessed from Android mobile devices, such as tablets and smartphones, either by downloading the APK file or by scanning the QR code from the MIT App Inventor application.

Finally, to control the robot's movement, we took advantage of the ESP32-CAM's ability to connect to a local Wi-Fi network . We configured the ESP32-CAM to act as a WebSocket server (a modern way of maintaining persistent connections between browser and server). Through this connection, we were able to send movement instructions, speed adjustments, and the activation of an additional light. Similarly, we received the images captured by the Rover's camera in real time.



7. Results

7.1. Demonstration video

https://youtu.be/_yPeMg6EOUQ?si=j1lwpJoroWA6bgUX

7.2. Materials and budget

Materiales	Precio
Tornillos M3 (diferentes tamaños)	S/ 22.5
Tuercas para tornillos M3	
Huacha plana tornillo 3mm	
Tuercas M6 con tuercas de seguridad	
Tornillos M6 45mm	
Varilla rosacda 8mm 285mm	
Tuerca M8	
Huacha plana tornillo 8mm	
Sierra	S/ 25.0
Soldador estaño y cautin	S/ 40.0
Legos	-
Cinta doble cara	S/ 5.0
Portapilas x2 pilas	S/ 7.5
Portapilas x4 pilas	S/ 14.0
Total	**S/ 114.0**

Componentes electrónicos	Precio
Motorreductor 200RPM 6V(con cable)	S/ 27
L298n motor driver	S/ 14
Baterías Litio 3.7 volteos	S/ 60
HC-05 Bluetooth module	S/ 30
HC-06 Bluetooth module	S/ 25
Sensor MQ-135	S/ 10
Sensor Ultrasonido	S/ 9
Sensor DTH11 (temperatura y humedad)	S/ 12
ESP32-CAM	S/ 65
Modulo GPS NEO-6M (gps)	S/ 45
Servo SG90	S/ 40
Cables macho-macho	S/ 15
Cables hembra-macho	
Cables hembra-hembra	
Total	**S/ 352**

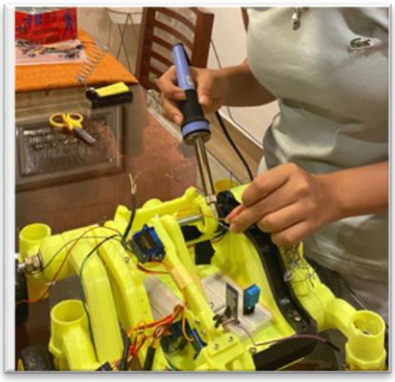
8. Challenges encountered

8.1. 3D Printing: We encountered difficulties with the adhesion of parts to the base, especially hollow parts, and we found it necessary to modify the STL files. We experimented with different types of supports and configurations on the 3D printer.

8.2. Assembly: At this stage, we identified that some parts in the pieces did not meet the correct dimensions, so we made manual adjustments using a drill, sandpaper, and hot needles to melt the plastic in small spaces.

8.3. Circuit construction: We tested with various power supplies and analyzed the power consumption of each sensor to avoid overloading. We encountered issues with the NEO-6M GPS module: it couldn't connect to any satellites, and when printing, it sent incorrect characters and not the desired longitude and latitude. Another challenge we faced was wire breakage when connecting them to the circuits, so we had to solder them multiple times.





8.4. Application programming: The complication arose with data transmission captured via Bluetooth. The sensor application sent information out of order and was updated late. In the robotic arm application, the slider adjustment did not match the requested degrees.

9. Conclusions

After finishing the construction of our *SeekerBot space prototype* , we came to the following conclusions:

- First, we were able to meet the proposed objectives. The rover successfully avoids obstacles, collects samples using its robotic claw, explores terrain with rough surfaces, sends information captured by sensors, and maintains a real-time connection with the user by sending images of its surroundings.

- Second, the prototype, at each stage, presented challenges that led us to explore solutions and expand our knowledge in robotics, electronics, design, and programming.

- Third, based on the experience gained, we propose for future projects: to consider the tolerance millimeters in the design of 3D parts, assemble the electronic circuits after the assembly of mechanical parts and perform partial functional tests during the process to identify and correct possible problems in time.

- Fourth, SeekerBot has the potential to be a valuable tool for space exploration and scientific research.